

2.0 Milestone Assessment: Network Systems

1. Introduction

Modern networks rely on multiple technologies like load balancing, redundancy (failover), and segmentation (subnetting/VLAN routing) to ensure high availability and performance. These technologies enable efficient data transmission by minimizing latency, avoiding congestion, and ensuring packets take the most optimal path between source and destination. Wireless configurations (e.g., channel selection, encryption like WPA3, and proper access point (**AP**) placement) further enhance connectivity by reducing interference and securing data in transit. As a network administrator, it is essential to understand how routing technologies, wireless configurations, and physical network installations each contribute to overall network functionality.

Networking can be viewed as a layered process in which data moves from physical signals, to structured communication, then on to intelligent path selection. Each layer depends on the one before/below it, meaning weaknesses in physical infrastructure or access configuration can directly impact routing efficiency and overall performance.

Routing serves as the decision-making system of the network, determining the best path for data to travel. However, it can only function effectively when supported by properly installed physical infrastructure and well-configured wired and wireless communication technologies.

To fully understand how these components interact, it is useful to follow how data moves through the network—from the physical layer, through network access, and ultimately to routing and delivery across networks.

2. Importance and Impact of Physical Network Installations

Physical network installations play a critical role in overall performance and reliability. Factors such as proper cable selection (Cat6 vs fiber), cable management, environmental controls (temperature/humidity), power protection (UPS), and physical security, directly impact uptime and signal quality. Poor installation, for instance, near electromagnetic interference (EMI), can lead to interference, packet loss, network downtime, and can disrupt network signals. Proper cable routing and shielding helps reduce these effects. A well-designed infrastructure supports scalability, consistent performance and helps mitigate the issues encountered.

The cable type and quality determine both bandwidth capacity and resistance to interference, with higher-grade cables such as Cat6 or fiber supporting faster and more reliable

communication. Considering distance limitations is a must, as Ethernet cables typically support a maximum length of 100 meters before signal degradation occurs.

Additionally, proper placement of networking devices, such as routers, switches, and APs, is critical and poor consideration can severely impact performance. For instance, AP dead zones or overheated equipment can reduce efficiency and reliability. Devices should be centrally located, properly ventilated, and free from physical obstructions.

Cable management and labeling are also important for maintenance and troubleshooting. Organized installations make it easier to identify issues and reduce downtime. Power considerations, including the use of uninterruptible power supplies (UPS), help maintain network availability during outages.

At this layer, data exists only as raw binary signals with no structure. If the physical layer is not properly designed or fails, communication across the network cannot occur. Overall, a well-designed physical installation ensures stable performance, reduces interference, and supports long-term scalability.

3. Wireless Data Transmission and Network Connectivity

The network access layer is responsible for how devices connect to and communicate within a local network. At this stage, raw signals are structured into frames and delivered using hardware-based addressing, Media Access Control (MAC) addresses.

Wired connections use Ethernet standards to provide stable, high-speed communication. Devices are connected through switches, which forward data frames based on MAC addresses. Wired networks offer low latency, minimal interference, and consistent performance, making them ideal for critical systems and backbone infrastructure.

Wireless standards developed by the IEEE, such as 802.11n (Wi-Fi 4, MIMO across 2.4/5 GHz to provide reliable, mid-range speed and coverage), 802.11ac (Wi-Fi 5, high-speed 5 GHz performance with wider channels for bandwidth-heavy applications), and 802.11ax (Wi-Fi 6, optimizes speed and capacity in crowded networks, supporting many devices efficiently), define performance capabilities and offer improvements in efficiency. Wireless configurations define the layout of devices and how they connect to a network without physical cables. Wireless networking extends connectivity using radio frequency signals, thereby allowing mobility and flexibility. It's important to keep in mind that proper configuration is essential to ensuring reliable performance, strong security, and user accessibility.

Key configuration factors include:

- **Service Set Identifier (SSID):** One of the primary components of wireless networking, defines the network name and should be clearly identified for users while avoiding unnecessary exposure.

- **Frequency Bands:** Wireless networks operate on different frequency bands, primarily 2.4 GHz and 5 GHz.
 - 2.4 GHz provides wider coverage but is more prone to interference and slower speeds.
 - 5 GHz offers higher speeds and reduced interference but has a shorter range.
- **Security Protocols:** Security is another critical factor. With WiFi Protected Access protocols, WPA2 and WPA3, encryption and protection against unauthorized access is provided. WPA3 is preferred for modern secure environments.
- **Channel Selection:** Plays a significant role in performance. Overlapping channels can cause interference, leading to slower speeds and unstable connections. Proper channel selection reduces interference from neighboring networks, improving performance and reliability.
- **Access Point Placement:** AP placement is crucial; devices should be centrally located, elevated, and free from obstructions to maximize coverage and signal strength.

By carefully selecting and configuring wireless technologies, network administrators can ensure reliable connectivity, strong performance, and secure access for users.

4. Routing Technologies, Path Selection and Delivery (IT532:OS2.1)

Routing technologies operate at the network layer and are responsible for directing data from a source device, between different networks, to its intended destination. Routers use logical addressing (IP addresses) and routing tables to determine the best path for data transmission.

These Routing tables store information about network destinations, next-hop addresses, and available paths. Routers evaluate these paths using metrics such as hop count, bandwidth, and delay to select the most efficient route.

Routing technologies are essential for directing traffic efficiently across networks. Key characteristics include: **dynamic routing protocols** (e.g., OSPF, BGP, RIP), which automatically learn and update routing information, adapts to changes in the network and is essential for modern, large-scale network environments; **static routing protocols**, which are suitable for simple, stable networks, provides manual control for predictable paths, but lacks flexibility; and path selection metrics such as cost, hop count, and bandwidth that determine the best route. Each is designed for different scales and purposes. Common dynamic routing protocols include:

1. **Open Shortest Path First (OSPF)** - a dynamic routing protocol used to automatically find the best path for data within a network. It is commonly used in large enterprise networks
 - a. dynamic, link-state routing protocol
 - b. Builds a full network map and selects the fastest path (based on cost)

- c. Fast convergence; ideal for large networks
 - d. Routers within the same area share a link-state database (LSDB)
- 2. **BGP (Border Gateway Protocol)** - the routing protocol used to exchange routing information between different networks (autonomous systems) - this protocol powers the internet.
 - a. Dynamic, path-vector routing protocol
 - b. Routes data between different networks using policies
 - c. Highly scalable; used across the internet
- 3. **RIP (Routing Information Protocol)** - one of the oldest dynamic routing protocols, mainly used in small or simple networks
 - a. Dynamic, distance-vector routing protocol
 - b. Chooses routes based on hop count (fewest routers)
 - c. Simple but slow; best for small networks

Another important feature is convergence, which refers to how quickly a network adapts to changes, such as a failed link. Faster convergence ensures minimal disruption to data flow, improves reliability and reduces downtime.

At this layer, data is packaged into packets containing source and destination IP addresses, allowing routers to determine where the data should be sent next. Overall, routing technologies contribute to efficient data transmission by optimizing paths, reducing delays, and maintaining connectivity, even when parts of the network may fail.

5. Interaction of Routing, Wireless, and Physical Technologies

Routing technologies, wireless configurations, and physical installations are interconnected and must perform as an integrated system in order to support overall network functionality. Each layer builds upon the previous one to enable complete communication.

When a user connects to a network, the Physical (Layer 1) infrastructure provides the pathways through which data travels. It transmits and receives signals through cables and hardware connections. The Data Link (Layer 2) organizes data into frames for delivery and wireless configurations enable user devices to access the network, acting as the entry point for many users. The Network (Layer 3) routing technologies then determine how data moves across the network and beyond, ensuring it reaches the correct destination efficiently.

For example, when a user connects to a wireless network:

- Their device communicates with an AP configured with proper security and channel settings.
- The AP connects through physical cabling to switches and routers.
- The routers then use routing protocols to forward the data along the best available path.

If any one of these components is misconfigured or fails, overall network performance is impacted. Together, these technologies create a cohesive system that supports reliable, secure, and efficient network operations.

6. Compare and contrast network access and management methods (IT532:OS2.3)

Network access and management methods define how devices connect to a network, how access is controlled, and how network devices are monitored and maintained. These functions primarily operate across OSI Layers 1 - 3, with a strong emphasis on the Physical (Layer 1) and Data Link (Layer 2), where devices initially connect and communicate within a local network.

Network access refers to the methods and technologies that allow devices to connect to a network and exchange data.

1. **Wired Access (Ethernet, Fiberoptic)** - Wired access uses Ethernet standards (IEEE 802.3) to connect devices through physical cabling. Communication at this level relies on MAC (Media Access Control) addresses, and data is transmitted in frames. Network switches use MAC address tables to forward traffic efficiently within the local network. Wired connections provide high reliability, low latency, and consistent performance, making them ideal for critical systems such as servers, network infrastructure devices, and desktop workstations.
2. **Wireless Access (Wi-Fi)** - Wireless access allows devices to connect using radio frequency signals based on IEEE 802.11 standards. Devices communicate through APs, which act as a bridge between wireless clients and the wired network. Effective wireless configuration requires consideration of several key factors: SSID (Service Set Identifier), Frequency Bands, Security Protocols, Channel Selection and AP Placement. Wireless networks provide flexibility and mobility, but performance can vary due to interference, signal strength, and environmental factors. Proper configuration is essential to maintain reliable connectivity.
3. **Secure Remote Access** - Remote access allows users to securely connect to a network from external locations using encrypted tunnels over the internet. Technologies such as IP Security (IPsec), Secure Socket Layer/Transport Layer Security (SSL/TLS) and virtual private networks (VPNs), protect data during transmission and enable secure access to internal network resources.

Network Access Control

Network access must be controlled to ensure only authorized users and devices can connect. This is typically implemented using Authentication, Authorization, and Accounting (AAA) principles:

- a. **Authentication:** Verifies the identity of a user or device

- b. Authorization: Determines what resources the user can access
- c. Accounting: Tracks user activity

Technologies such as RADIUS and IEEE 802.1X are commonly used to enforce secure access control in both wired and wireless networks.

Network Management Methods

Network management methods are used to configure, monitor, and maintain network devices to ensure optimal performance and security.

1. **Local Management** - involves direct access to a device, typically through a console connection. This method is commonly used for initial setup or troubleshooting when remote access is unavailable.
2. **Remote Management** - allows administrators to configure devices over the network using secure protocols such as SSH or web-based interfaces (HTTPS). This enables efficient management without requiring physical access to devices.
3. **Centralized Management Systems** - allow administrators to manage multiple devices from a single platform. Examples include wireless controllers that manage multiple APs and network management systems (NMS) that provide visibility and control across the network.
4. **Network Monitoring Tools** - tools that track network performance, device status, and traffic patterns. Protocols such as Simple Network Management Protocol (SNMP) are commonly used to collect data and identify potential issues before they impact users.

Summary

Network access and management methods are essential components of modern network design. Wired, wireless, and remote access technologies enable connectivity, while access control mechanisms ensure security. Management methods provide the tools necessary to maintain performance, monitor activity, and scale the network effectively.

By properly configuring access methods and implementing effective management practices, network administrators can ensure that routing technologies operate on reliable connections.

Together these elements support reliable communication and directly contribute to the efficiency and effectiveness of routing technologies within the network. The results are improved performance, reduced downtime, and more efficient data delivery across networks.

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